Matrix operations and FFT

Begin with block matrices and multiplication, proceed to elimination, PA=LU and least squares, then use the DFT and FFT for polynomial convolution.

Matrix methods organize computation through blocks and factorizations; FFT reuses roots of unity through even-odd splitting. Track dimensions, pivots, transform conventions and padding.

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Reuse the same coefficient matrix

In $Ax = b$, separate the coefficients from the right-hand side. Reuse a factorization when A stays fixed and b changes. Use real matrices, column vectors, $PA = LU$, and unit-diagonal L. Define the least-squares residual as $r = b - A\hat{x}$.

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Match the inner dimensions

For $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{k \times n}$, $C = AB \in \mathbb{R}^{m \times n}$ with $c_{ij} = \sum_{t=1}^{k} a_{it} b_{tj}$. The direct algorithm uses mkn multiplications and $mn(k-1)$ additions. A defined AB does not imply a defined BA.

Matrix dimensions and one dot product

A: m × k

B: k × n

AB: m × n

Shared index t = 1…k

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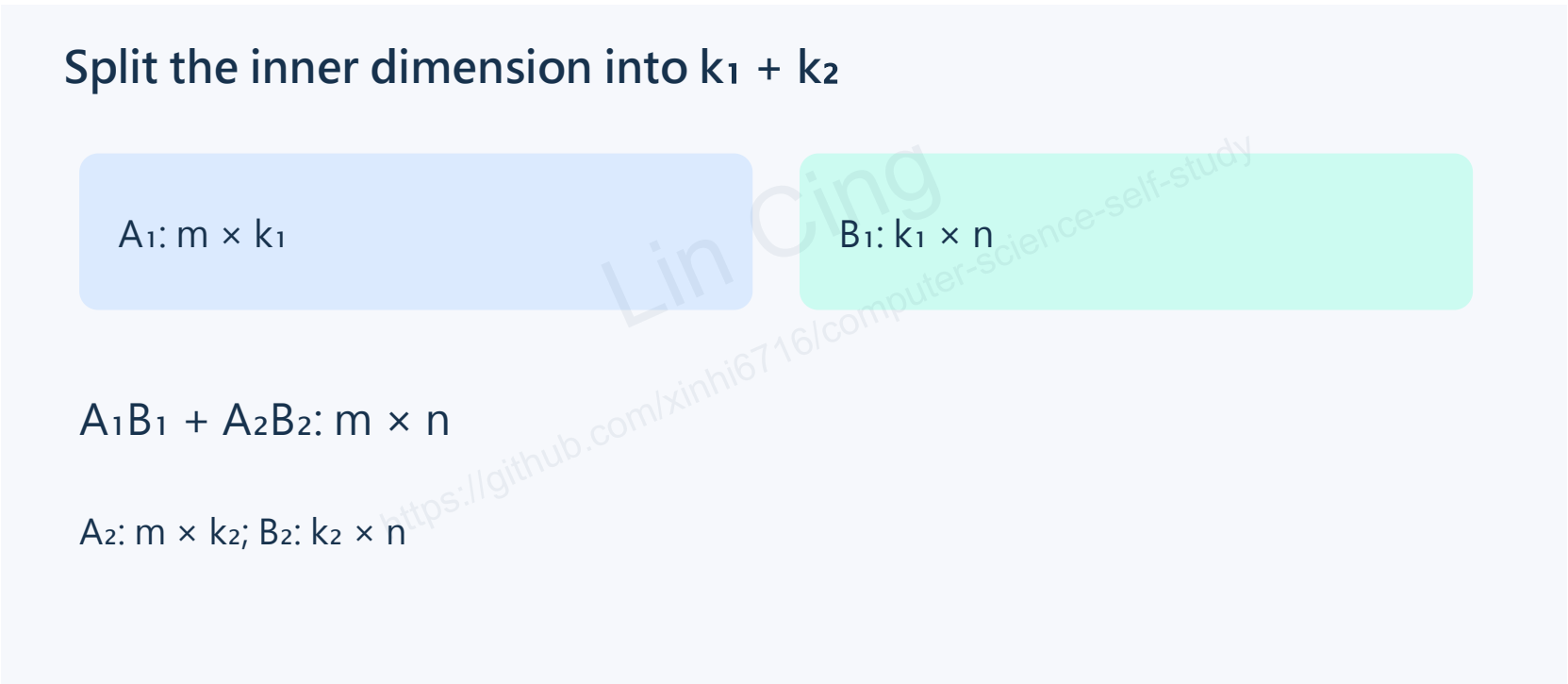
Compute entries with their indices

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. Then $c_{12} = 1(0) + 2(1) + 3(1) = 5$ and $AB = \begin{bmatrix} 4 & 5 \\ 10 & 11 \end{bmatrix}$. Here BA is 3×3, so it cannot equal AB.

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Check dimensions inside every block

Split the inner dimension as $k = k_1 + k_2$. For $A = [A_1\ A_2]$ and $B = [B_1; B_2]$, let A_s have size $m \times k_s$ and B_s have size $k_s \times n$. Then $AB = A_1B_1 + A_2B_2$. In a two-by-two partition, $C_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j}$. Blocks need not commute.



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Fewer recursive products change the exponent

Ordinary square blocking gives $8T(n/2) + \Theta(n^2) = \Theta(n^3)$. Strassen reconstructs four output blocks from seven products, giving $7T(n/2) + \Theta(n^2) = \Theta(n^{\log_2 7})$. Pad with zeros to a power of two and crop the result. Arithmetic complexity does not guarantee faster small instances.

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Strassen: form seven products

Partition square matrices into equal blocks $A = [A_{11}, A_{12}; A_{21}, A_{22}]$ and $B = [B_{11}, B_{12}; B_{21}, B_{22}]$. Form seven block products:

$P_1 = (A_{11} + A_{22})(B_{11} + B_{22})$; $P_2 = (A_{21} + A_{22})B_{11}$; $P_3 = A_{11}(B_{12} - B_{22})$; $P_4 = A_{22}(B_{21} - B_{11})$.

$P_5 = (A_{11} + A_{12})B_{22}$; $P_6 = (A_{21} - A_{11})(B_{11} + B_{12})$; $P_7 = (A_{12} - A_{22})(B_{21} + B_{22})$. Block multiplication is not commutative.

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Strassen: recombination and proof

$C_{11} = P_1 + P_4 - P_5 + P_7, C_{12} = P_3 + P_5; C_{21} = P_2 + P_4, C_{22} = P_1 - P_2 + P_3 + P_6.$

For instance, expanding C_{11} cancels the cross terms and leaves $A_{11}B_{11} + A_{12}B_{21}$. Expanding the other blocks gives ordinary block multiplication. Seven half-size products and $\Theta(n^2)$ additions per level yield $T(n) = 7T(n/2) + \Theta(n^2) = \Theta(n^{\log_2 7})$; zero-pad to a power of two if needed.

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Strassen: a 2×2 trace

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. The seven scalar products are $P_1 = 65$, $P_2 = 35$, $P_3 = -2$, $P_4 = 8$, $P_5 = 24$, $P_6 = 22$, $P_7 = -30$.

Recombination gives $C_{11} = 65 + 8 - 24 - 30 = 19$, $C_{12} = -2 + 24 = 22$, $C_{21} = 35 + 8 = 43$, and $C_{22} = 65 - 35 - 2 + 22 = 50$, matching direct multiplication. Arithmetic counts alone do not capture finite precision or small-size constants.

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Elimination stores multipliers and updates the tail

At step k, use a nonzero pivot u_{kk}, set $l_{ik} = a_{ik}/u_{kk}$, and apply $R_i \leftarrow R_i - l_{ik}R_k$. The surviving upper triangle is U; the multipliers form L. The first block update is the Schur complement $A_{22} - a_{21}a_{12}/a_{11}$.

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An invertible matrix can have a zero pivot

The matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ has determinant -1, but elimination without a row exchange divides by zero immediately. With $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = IU$. A zero chosen pivot alone does not prove that the original matrix is singular.

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First-column row exchange and elimination

Use the CLRS28.1 example $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 4 \\ 5 & 6 & 3 \end{bmatrix}$. Partial pivoting chooses the largest available absolute entry.

Swap $R_1 \leftrightarrow R_3$ to obtain $[5, 6, 3; 3, 4, 4; 1, 2, 0]$. With multipliers $3/5$ and $1/5$, eliminate to obtain $[5, 6, 3; 0, 2/5, 11/5; 0, 4/5, -3/5]$.

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First-column row exchange and elimination (continued)

Round 1: compare, swap, eliminate

A

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 4 \\ 5 & 6 & 3 \end{bmatrix}$$



P_1A

$$\begin{bmatrix} 5 & 6 & 3 \\ 3 & 4 & 4 \\ 1 & 2 & 0 \end{bmatrix}$$

U after round 1

$$\begin{bmatrix} 5 & 6 & 3 \\ 0 & 2/5 & 11/5 \\ 0 & 4/5 & -3/5 \end{bmatrix}$$

$|1| < |3| < |5| \rightarrow R_1 \leftrightarrow R_3$

$l_{21} = 3/5, l_{31} = 1/5; R_2 \leftarrow R_2 - (3/5)R_1; R_3 \leftarrow R_3 - (1/5)R_1$

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A later swap must move stored multipliers

In column two, $4/5 > 2/5$, so exchange working rows two and three. Also exchange the already stored first-column multipliers in L, giving $l_{21} = 1/5, l_{31} = 3/5$. The working matrix is $[5, 6, 3; 0, 4/5, -3/5; 0, 2/5, 11/5]$. With $l_{32} = 1/2$, the final pivot is $11/5 - (1/2)(-3/5) = 5/2$.

Round 2: swap the working matrix, P and stored L

U_1

$$\begin{bmatrix} 5 & 6 & 3 \\ 0 & 2/5 & 11/5 \\ 0 & 4/5 & -3/5 \end{bmatrix}$$

P_1

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

L: stored column 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 3/5 & 1 & 0 \\ 1/5 & 0 & 1 \end{bmatrix}$$

$|4/5| > |2/5| \rightarrow R_2 \leftrightarrow R_3$

Before the swap; exchange only the completed column of L.

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Verify the convention with PA=LU

$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 & 0 \\ 1/5 & 1 & 0 \\ 3/5 & 1/2 & 1 \end{bmatrix}$, and $U = \begin{bmatrix} 5 & 6 & 3 \\ 0 & 4/5 & -3/5 \\ 0 & 0 & 5/2 \end{bmatrix}$. Both PA and LU equal $[5, 6, 3; 1, 2, 0; 3, 4, 4]$. P reorders equations, leaving the unknown ordering unchanged.

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Verify the convention with PA=LU (continued)

Eliminate and verify every entry of PA = LU

U after $R_2 \leftrightarrow R_3$

$$\begin{bmatrix} 5 & 6 & 3 \\ 0 & 4/5 & -3/5 \\ 0 & 2/5 & 11/5 \end{bmatrix}$$

U

$$\begin{bmatrix} 5 & 6 & 3 \\ 0 & 4/5 & -3/5 \\ 0 & 0 & 5/2 \end{bmatrix}$$

PA = LU

$$\begin{bmatrix} 5 & 6 & 3 \\ 1 & 2 & 0 \\ 3 & 4 & 4 \end{bmatrix}$$

$l_{32} = (2/5)/(4/5) = 1/2$; $R_3 \leftarrow R_3 - (1/2)R_2$

LU rows: (5, 6, 3); (1, 6/5+4/5, 3/5−3/5);

(3, 18/5+2/5, 9/5−3/10+5/2) = (3, 4, 4)

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Why can LUP keep finding a pivot?

Consider an invertible current trailing block K. Its first column contains a nonzero entry; otherwise K would have a zero column and be singular. Swap rows within the trailing block to move such an entry to the pivot a. The swap preserves invertibility.

Write $K' = PK$ for the block after the swap. Block elimination gives $\det K' = a \det S$, while $\det K' = \pm \det K$. Since $\det K \neq 0$ and $a \neq 0$, the new Schur complement S is invertible, and induction continues to the last pivot. Later row swaps also exchange corresponding multipliers already stored in L, producing $PA = LU$. Selecting the largest absolute pivot controls local multipliers but does not guarantee numerical stability for every problem.

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In-place LUP: inputs, outputs and storage

Input is a real $n \times n$ matrix A. Start with working array $W = A$ and permutation vector $p = [1, \dots, n]$. On success, the diagonal and upper triangle of W store U, its strict lower triangle stores the multipliers of L, and the unit diagonal of L is implicit. The convention is $(Pb)_i = b_{p_i}$ and $PA = LU$.

The algorithm below uses exact arithmetic and one-based indices; for $n = 0$ one may return the empty factorization. "In place" means overwriting the working matrix; preserving the original A requires another $\Theta(n^2)$ storage.

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In-place LUP: pivot, swap the whole row, update

For $k = 1, \ldots, n$, choose q maximizing $|W_{ik}|$ among $i = k, \ldots, n$. If $W_{qk} = 0$, report a singular matrix and stop; otherwise exchange entire rows k, q of W, and exchange p_k, p_q.

For each $i = k + 1, \ldots, n$, first store $W_{ik} \leftarrow W_{ik}/W_{kk}$; then for $j = k + 1, \ldots, n$ update $W_{ij} \leftarrow W_{ij} - W_{ik}W_{kj}$. Swapping whole rows includes earlier multipliers in columns $j < k$. Swapping only the trailing block would break $PA = LU$.

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Invariant for the completed blocks

After the first k steps, the currently permuted original matrix satisfies

$PA = \begin{bmatrix} L_{11} & 0 \\ L_{21} & I \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} \\ 0 & S \end{bmatrix}$, with $k \times k$ leading triangular blocks and remaining Schur complement S.

Completed columns L_{11}, L_{21} and completed rows U_{11}, U_{12} are stored in W.

The next row swap also reorders the already-stored multipliers in L_{21}. Dividing below the new pivot gives the new multipliers, and the outer-product update modifies S exactly as block multiplication requires.

Initially $k = 0$ and $S = A$; at $k = n$ the trailing block is empty and $PA = LU$.

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What a zero pivot reports

Under exact arithmetic, if all remaining entries in the current pivot column are zero, the trailing block has a zero column and is singular. Earlier pivots were nonzero and all previous permutations and elimination operations were invertible, so the original A is singular. Conversely, an invertible A has an invertible trailing block at each step and hence a nonzero candidate pivot.

In floating point, treating a small pivot as zero under a chosen scale and tolerance only reports failure under that rule; it does not prove exact mathematical singularity. Rounding, scaling and conditioning matter. Numerical rank requires an appropriate rank-revealing method and an explicit tolerance.

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Trace the in-place array

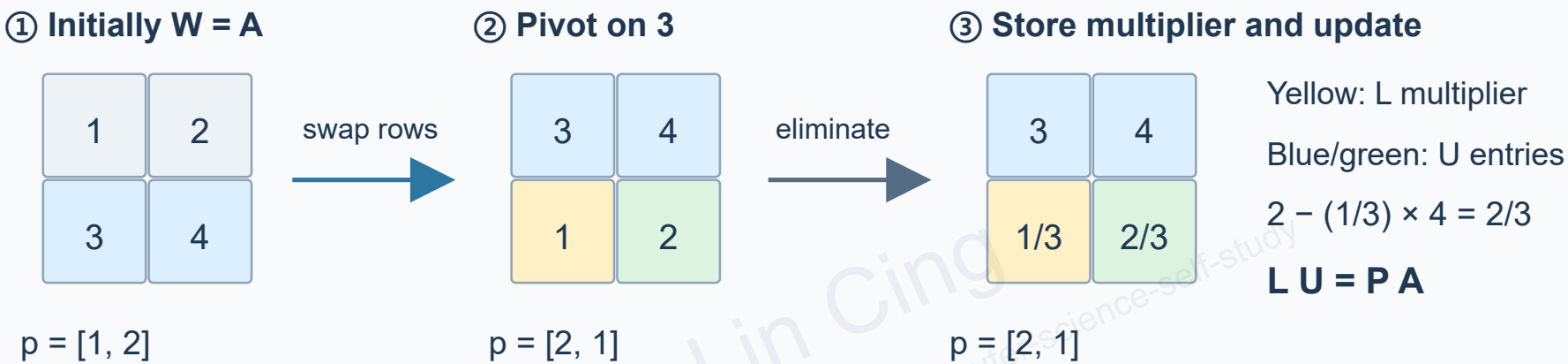
Take $A = [1, 2; 3, 4]$ and $b = (5, 11)^T$. The largest absolute value in column 1 is 3. Swap rows and set $p = (2, 1)$: the working matrix becomes $[3, 4; 1, 2]$.

Compute $l_{21} = 1/3$ and update the lower-right entry to $2 - (1/3)4 = 2/3$. Thus $W = [3, 4; 1/3, 2/3]$ stores $L = [1, 0; 1/3, 1]$ and $U = [3, 4; 0, 2/3]$. Check $LU = [3, 4; 1, 2] = PA$ and $Pb = (11, 5)^T$.

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Trace the in-place array (continued)

In-place LUP: one working array at three instants



Swap rows including stored multipliers; this first step has no prior multipliers.

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Forward substitution computes the intermediate vector

For $b = (3, 7, 8)^T$, permute to $Pb = (8, 3, 7)^T$. Solve $Ly = Pb$. Then $y_1 = 8$, $y_2 = 3 - (1/5)8 = 7/5$, and $y_3 = 7 - (3/5)8 - (1/2)(7/5) = 3/2$. In general, $y_i = (Pb)_i - \sum_{j<i} l_{ij} y_j$. One forward solve takes $\Theta(n^2)$ time.

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Back substitution starts at the last row

Solve $Ux = y$. Obtain $x_3 = (3/2)/(5/2) = 3/5$, $x_2 = [7/5 + (3/5)(3/5)]/(4/5) = 11/5$, and $x_1 = [8 - 6(11/5) - 3(3/5)]/5 = -7/5$. Substitution gives $Ax = (3, 7, 8)^T$. In general, $x_i = (y_i - \sum_{j>i} u_{ij} x_j)/u_{ii}$.

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Factor once for many right-hand sides

Dense elimination updates $\sum_{k=1}^{n-1}(n-k)^2 = \Theta(n^3)$ trailing entries. With the factors available, each new right-hand side needs two triangular solves in $\Theta(n^2)$. For q right-hand sides, the total is $\Theta(n^3 + qn^2)$. Partial pivoting can still suffer element growth; it does not guarantee accuracy for every problem.

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Derive determinants and inverses from the factors

After s row exchanges, $\det A = (-1)^s \prod_i u_{ii}$. Two exchanges give $\det A = 5(4/5)(5/2) = 10$. To invert, solve $Az_j = e_j$; z_j is inverse column j. After factorization, n solves cost $\Theta(n^3)$. For a single $Ax = b$, use the factors directly instead of computing the whole inverse.

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Forward and back substitution invariants

For unit lower-triangular L, solve $Ly = Pb$ by $y_i = (Pb)_i - \sum_{j<i} l_{ij} y_j$. Before step i, rows $1, \ldots, i-1$ hold and those y_j stay fixed. The formula makes row i hold; induction gives $Ly = Pb$.

For nonsingular upper-triangular U, solve $Ux = y$ from $i = n$ down to 1 using $x_i = (y_i - \sum_{j>i} u_{ij} x_j)/u_{ii}$. Before step i, rows $i+1, \ldots, n$ hold; their equations do not depend on x_i, and the step establishes row i. Nonzero pivots make division valid.

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Both triangular solves on the same example

Forward substitution gives $y_1 = 11$ and $y_2 = 5 - (1/3)11 = 4/3$. Back substitution gives $x_2 = (4/3)/(2/3) = 2$ and $x_1 = (11 - 4 \cdot 2)/3 = 1$. Finally $A(1, 2)^T = (5, 11)^T = b$.

Row 2 of the upper-triangular system holds first. Computing x_1 leaves x_2 unchanged, so row 2 still holds: this is the back-substitution invariant in a concrete calculation.

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Block inversion starts with elimination

Let $A = \begin{bmatrix} B & C \\ D & E \end{bmatrix}$, with square, invertible B and compatible dimensions. Its Schur complement is $S = E - DB^{-1}C$. Block elimination gives

$$\begin{bmatrix} I & 0 \\ -DB^{-1} & I \end{bmatrix} A = \begin{bmatrix} B & C \\ 0 & S \end{bmatrix}.$$

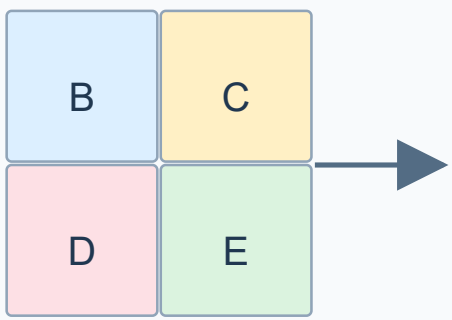
Eliminate the lower-left block first, then work on S. Block products cannot be reordered.

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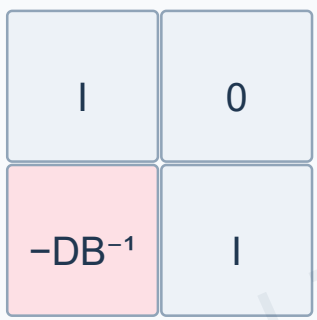
Block inversion starts with elimination (continued)

Schur complement: eliminate lower left and update lower right

Original matrix A

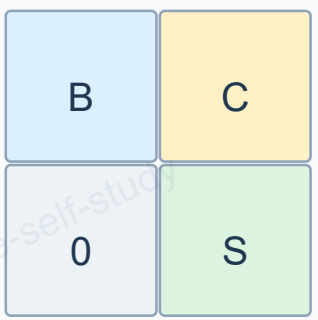


Left-multiply by elimination matrix



× A

Resulting upper block triangle



Lower-right S is the updated trailing block

If B is invertible, $S = E - DB^{-1}C$; A is invertible iff S is.

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Block inverse and its hypotheses

If both B and S are invertible, reversing the block elimination factors yields

$$A^{-1} = \begin{bmatrix} B^{-1} + B^{-1}CS^{-1}DB^{-1} & -B^{-1}CS^{-1} \\ -S^{-1}DB^{-1} & S^{-1} \end{bmatrix}.$$

Check the result by multiplying AA^{-1}. When B is invertible, A is invertible exactly when S is. If B is singular this formula does not apply, even though A itself may be invertible.

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A block inverse: compute the Schur complement

Take $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 5 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and partition after the first two rows and columns. Then $B = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$, $C = (2,1)^T$, $D = C^T$, $E = 3$.

$B^{-1} = \frac{1}{16} \begin{bmatrix} 5 & -2 \\ -2 & 4 \end{bmatrix}$, $B^{-1}C = (1/2, 0)^T$, and $S = 3 - (2,1)(1/2,0)^T = 2$.

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Assemble the inverse block by block

$S^{-1} = 1/2$ and the upper-right block is $-B^{-1}CS^{-1} = (-1/4, 0)^T$; the lower-left block is its transpose.

The upper-left block is $B^{-1} + (1/2, 0)^T (1/2)(1/2, 0) = \begin{bmatrix} 7/16 & -1/8 \\ -1/8 & 1/4 \end{bmatrix}$.

Thus $A^{-1} = \begin{bmatrix} 7/16 & -1/8 & -1/4 \\ -1/8 & 1/4 & 0 \\ -1/4 & 0 & 1/2 \end{bmatrix}$. For example, the first row times the first column gives $4(7/16) + 2(-1/8) + 2(-1/4) = 1$.

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Positive definiteness protects recursive pivots

A real symmetric matrix is positive definite (SPD) when $z^T Az > 0$ for every nonzero z. With $A = \begin{bmatrix} B & C \\ C^T & D \end{bmatrix}$, the principal block B is SPD: choose $z = (x, 0)$ for nonzero x.

The Schur complement $S = D - C^T B^{-1} C$ is also SPD: for nonzero y choose $z = (-B^{-1}Cy, y)$, so $y^T Sy = z^T Az > 0$. Consequently B and S are invertible and the block formula can recurse.

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Cholesky: square root and trailing update

Every real symmetric positive-definite matrix has a unique factorization $A = LL^T$, where L is lower triangular with positive diagonal. Partition $A = \begin{bmatrix} a & r^T \\ r & D \end{bmatrix}$. Since $a > 0$, set $\ell_{11} = \sqrt{a}$ and $v = r/\sqrt{a}$; the remaining matrix $S = D - vv^T$ is SPD.

Recursively factor $S = L_2 L_2^T$ to obtain $L = \begin{bmatrix} \sqrt{a} & 0 \\ v & L_2 \end{bmatrix}$. Each square root is positive. The first column is forced by the positive diagonal convention, and induction proves uniqueness.

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Elementwise Cholesky recurrence

For real symmetric positive-definite $A = LL^T$ with positive-diagonal lower-triangular L, the identity $a_{ij} = \sum_{k=1}^{\min(i,j)} l_{ik} l_{jk}$ gives the column-by-column recurrence for $j = 1, \ldots, n$:

$l_{jj} = \sqrt{a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2}$; for $i > j$, $l_{ij} = \dfrac{a_{ij} - \sum_{k=1}^{j-1} l_{ik} l_{jk}}{l_{jj}}$.

An empty sum is zero. Compute the diagonal first, then the entries below it; all columns $k < j$ have already been computed. Entries above the diagonal are zero.

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Why are the square roots and divisions valid?

At step j, the radicand is the leading diagonal entry of the Schur complement remaining after eliminating columns $1, \ldots, j-1$. Principal blocks and Schur complements of an SPD matrix are SPD, so this entry is strictly positive. Thus $l_{jj} > 0$ and the division below it is defined.

Substituting the computed entries into $a_{ij} = \sum_{k \leq \min(i,j)} l_{ik} l_{jk}$ shows, one column at a time, that $A = LL^T$. The positive diagonal forces each square root and each entry below it uniquely. Without real symmetry and positive definiteness, the radicand or divisor need not be positive.

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Compute every entry in the 3×3 example

For $A = [4, 2, 2; 2, 5, 1; 2, 1, 3]$, the first column is $l_{11} = \sqrt{4} = 2$, $l_{21} = 2/2 = 1$, $l_{31} = 2/2 = 1$.

The second column is $l_{22} = \sqrt{5 - l_{21}^2} = 2$, $l_{32} = (1 - l_{31} l_{21})/l_{22} = 0$. Finally $l_{33} = \sqrt{3 - l_{31}^2 - l_{32}^2} = \sqrt{2}$.

Thus $L = [2, 0, 0; 1, 2, 0; 1, 0, \sqrt{2}]$. Check $l_{31} l_{21} + l_{32} l_{22} = 1 = a_{32}$ and verify every entry with $LL^T = A$.

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Cholesky example: follow the update

For $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 5 & 1 \\ 2 & 1 & 3 \end{bmatrix}$, the first pivot is $\ell_{11} = 2$, and both entries below it equal 1. The trailing update is $\begin{bmatrix} 5 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$.

Hence $L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & \sqrt{2} \end{bmatrix}$. Multiplication gives $LL^T = A$. Also $x^T Ax = \|L^T x\|^2 > 0$ for nonzero x, verifying that this example is SPD.

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Cholesky example: follow the update (continued)

Cholesky: update the trailing block from the first pivot

Matrix A

4	2	2
2	5	1
2	1	3

Blue: first column

Green: trailing block

1 Compute first column

$\ell_{11} = \sqrt{4} = 2$

$v = [2/2, 2/2]^T = [1, 1]^T$

2 Subtract each corresponding product

$$\begin{bmatrix} 5 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 1\cdot1 & 1\cdot1 \\ 1\cdot1 & 1\cdot1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

For example: lower right $3 - 1\cdot1 = 2$

3 Factor the new block: $\sqrt{4} = 2$ and $\sqrt{2}$

$L = [2, 0, 0; 1, 2, 0; 1, 0, \sqrt{2}]$ $LL^T = A$

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Solve using a Cholesky factor

Solve $LL^T x = b$ by forward substitution $Ly = b$ and back substitution $L^T x = y$. For $b = (8, 8, 6)^T$, forward substitution gives $y = (4, 2, \sqrt{2})^T$. Back substitution yields $x_3 = 1$, $x_2 = 1$, $x_1 = 1$; check $A(1, 1, 1)^T = b$.

Dense Cholesky factorization costs $\Theta(n^3)$ arithmetic operations; each subsequent right-hand side costs $\Theta(n^2)$ for both triangular solves. The SPD hypothesis matters; near-positive-definite floating-point inputs also need rounding and failure handling.

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Matrix inversion can compute a product

For two $n \times n$ matrices X, Y, form $T = \begin{bmatrix} I & X & 0 \\ 0 & I & Y \\ 0 & 0 & I \end{bmatrix}$.

$T^{-1} = \begin{bmatrix} I & -X & XY \\ 0 & I & -Y \\ 0 & 0 & I \end{bmatrix}$. Block multiplication checks the formula: the upper-right block is $XY - XY = 0$ and the other off-diagonal blocks cancel. Read XY from the upper-right block; T is invertible for every X, Y.

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Matrix inversion can compute a product (continued)

Read a matrix product from the upper-right inverse block

Input: arbitrary X and Y

$$T = \begin{bmatrix} I & X & 0 \\ 0 & I & Y \\ 0 & 0 & I \end{bmatrix}$$

invert

Output: read the green block

$$T^{-1} = \begin{bmatrix} I & -X & XY \\ 0 & I & -Y \\ 0 & 0 & I \end{bmatrix}$$

Green block = XY

Dimensions: n × n

Example: X = [2], Y = [3]

Upper-right block = [6]

Build T: $O(n^2)$; assume $I(3n)=O(I(n))$

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The cost of extracting a product

Let $I(n)$ bound the cost of inverting an invertible $n \times n$ matrix. If $I(n) = \Omega(n^2)$ and $I(3n) = O(I(n))$, forming the $3n \times 3n$ matrix T and extracting its upper-right block takes $O(n^2)$, so matrix multiplication costs $M(n) = O(I(n))$.

This reduction counts arithmetic operations. The scaling assumption matters: a bound for one dimension alone does not justify replacing $I(3n)$ by $I(n)$.

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Fast multiplication gives fast inversion

For an SPD matrix, the block formula recursively inverts B and S, and uses a constant number of half-size products and $O(n^2)$ additions. Thus $I_{SPD}(n) \leq 2I_{SPD}(n/2) + O(M(n) + n^2)$.

If $M(n) = \Omega(n^2)$ and $M(n/2) \leq cM(n)$ for a fixed $c < 1/2$, successive recursion levels cost at most a ratio $2c < 1$ of the preceding bound. Summing yields $I_{SPD}(n) = O(M(n))$. The conclusion needs these cost assumptions.

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General invertible matrices and the arithmetic model

For real nonsingular A, A^TA is SPD since $x^TA^TAx = ||Ax||^2 > 0$ for nonzero x. Form and invert A^TA, then multiply by A^T: $A^{-1} = (A^TA)^{-1}A^T$. To pad to a power of two, use $\text{diag}(A, I)$ and assume $M(n+k) = O(M(n))$ for $0 \leq k \leq n$.

Under the stated regularity conditions and exact arithmetic model, multiplication and inversion reduce to one another. This does not guarantee floating-point stability: forming A^TA squares the condition number, so practical solves typically use pivoted elimination or a suitable factorization.

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The relation between LDL^T and Cholesky

For SPD matrices we may retain the elimination pivots in a diagonal matrix: $A = LDL^T$, where L is unit lower triangular and D is positive diagonal. Each pivot is the top-left entry of the current SPD Schur complement and is therefore positive.

For $D = \mathrm{diag}(d_1, \ldots, d_n)$, set $D^{1/2} = \mathrm{diag}(\sqrt{d_i})$. Then $A = (LD^{1/2})(LD^{1/2})^T$, giving the positive-diagonal Cholesky factor. In the previous example $L = [1, 0, 0; 1/2, 1, 0; 1/2, 0, 1]$, $D = \mathrm{diag}(4, 4, 2)$; multiplying by $D^{1/2}$ gives $[2, 0, 0; 1, 2, 0; 1, 0, \sqrt{2}]$.

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Leading arithmetic cost of dense factorizations

Assume dense $n \times n$ input, count each scalar addition, subtraction, multiplication and division as one flop, and no early singular return. LUP trailing updates contribute about $2\sum_{s=1}^{n-1} s^2 = \frac{2}{3}n^3 + O(n^2)$ flops; multiplier divisions add $O(n^2)$. The two triangular solves for one right-hand side take about $2n^2 + O(n)$ flops together.

For Cholesky, elementwise multiply-add terms take $\frac{1}{3}n^3 + O(n^2)$ flops; divisions and square roots occur $O(n^2)$ and n times respectively. Square roots may have a different cost and should be reported separately. These leading coefficients depend on the stated flop convention and are not wall-clock-time ratios.

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Account separately for pivots and memory traffic

Partial pivot selection scans absolute values and makes $O(n^2)$ comparisons. At most n full-row swaps move $O(n^2)$ matrix entries and $O(n)$ permutation indices. These comparisons, memory accesses and square roots are not normally included in the multiply-add flop totals; actual time can be dominated by memory behavior or implementation choices.

W occupies $\Theta(n^2)$ storage and the permutation needs $\Theta(n)$ extra space. Whether the original matrix is retained changes the additional storage. Sparse matrices, low-rank shortcuts, parallelism and early exits are outside the direct dense full-run time accounting.

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Three noncollinear points require a fit

Fit $b \approx \alpha + \beta t$ to $(t, b) = (0, 1), (1, 2), (2, 2)$. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$, $x = (\alpha, \beta)^T$, and $b = (1, 2, 2)^T$.

Consecutive slopes are 1 and 0, so no line passes through all three points. Minimize $||b - Ax||_2^2$.

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Set the gradient to zero for the normal equations

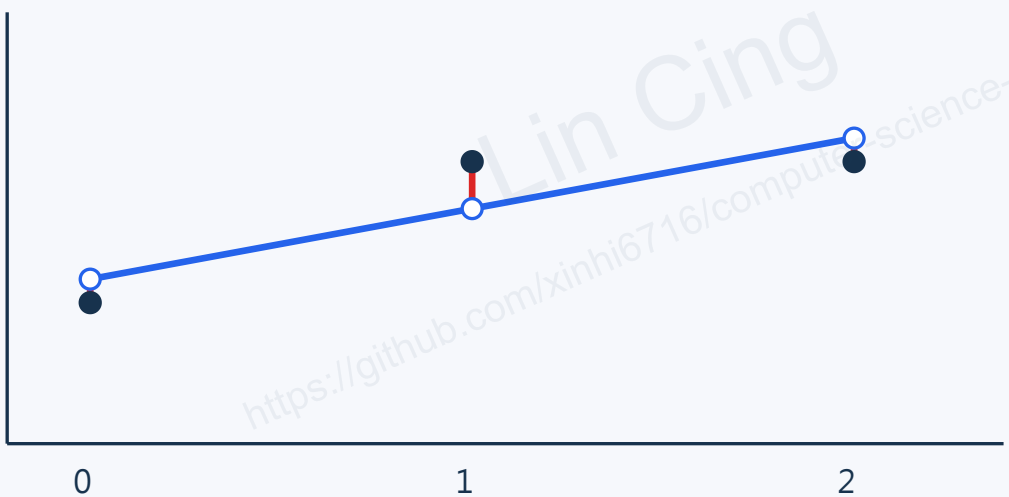
The gradient is $-2A^T(b - Ax)$, so $A^T A\hat{x} = A^T b$. Here $A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$ and $A^T b = (5, 6)^T$. Full column rank gives $z^T A^T Az = \|Az\|^2 > 0$ for every nonzero z, hence a unique solution. Dependent columns invalidate that uniqueness argument.

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Compute coefficients, predictions, and residuals

The equations are $3\alpha + 3\beta = 5$ and $3\alpha + 5\beta = 6$. Subtraction gives $\beta = 1/2$, then $\alpha = 7/6$. Predictions are $A\hat{x} = (7/6, 5/3, 13/6)^T$ and residuals are $r = (-1/6, 1/3, -1/6)^T$. The squared error is $1/36 + 1/9 + 1/36 = 1/6$.

Vertical residuals, not shortest point-line distances



$b = (1,2,2)^T$

$\hat{b} = (7/6,5/3,13/6)^T$

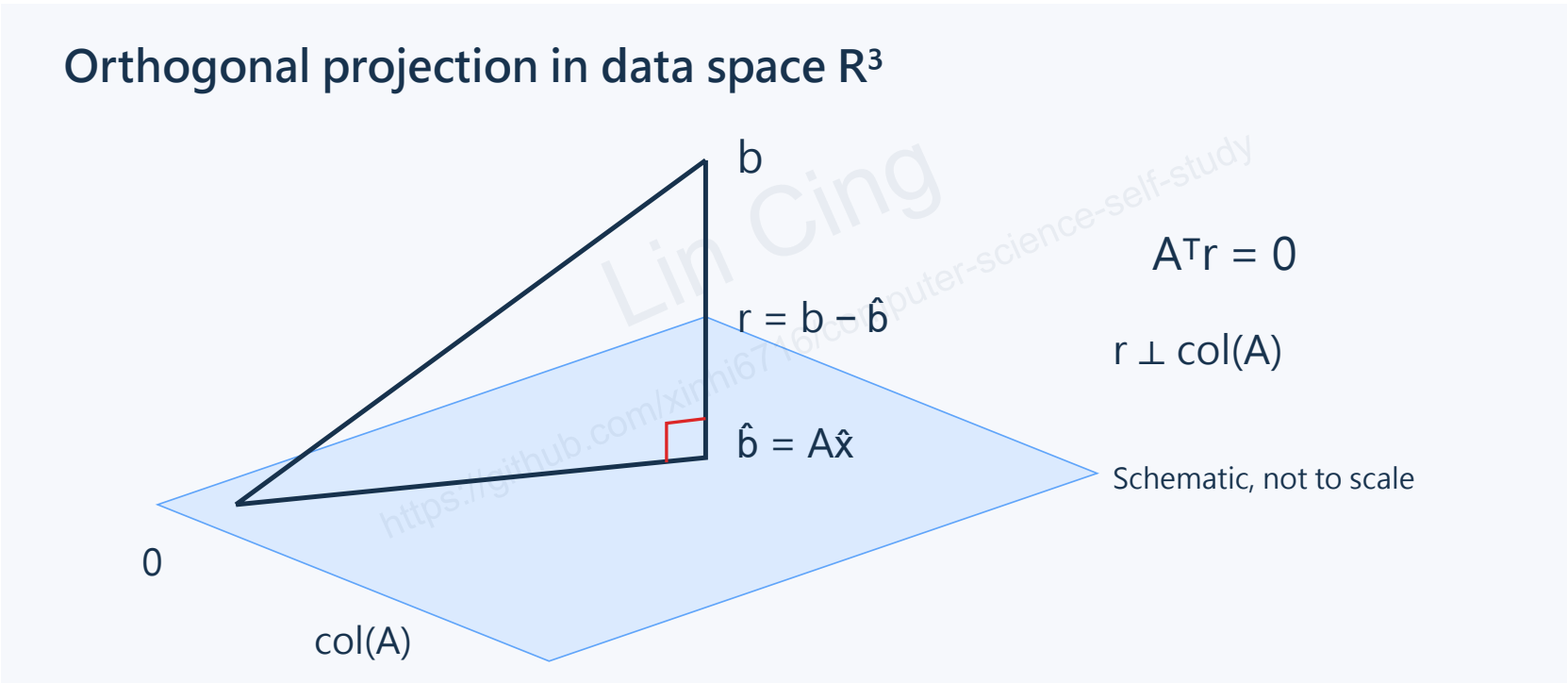
$r = (-1/6,1/3,-1/6)^T$

Red segments fix t; blue line $\hat{b} = 7/6 + t/2$

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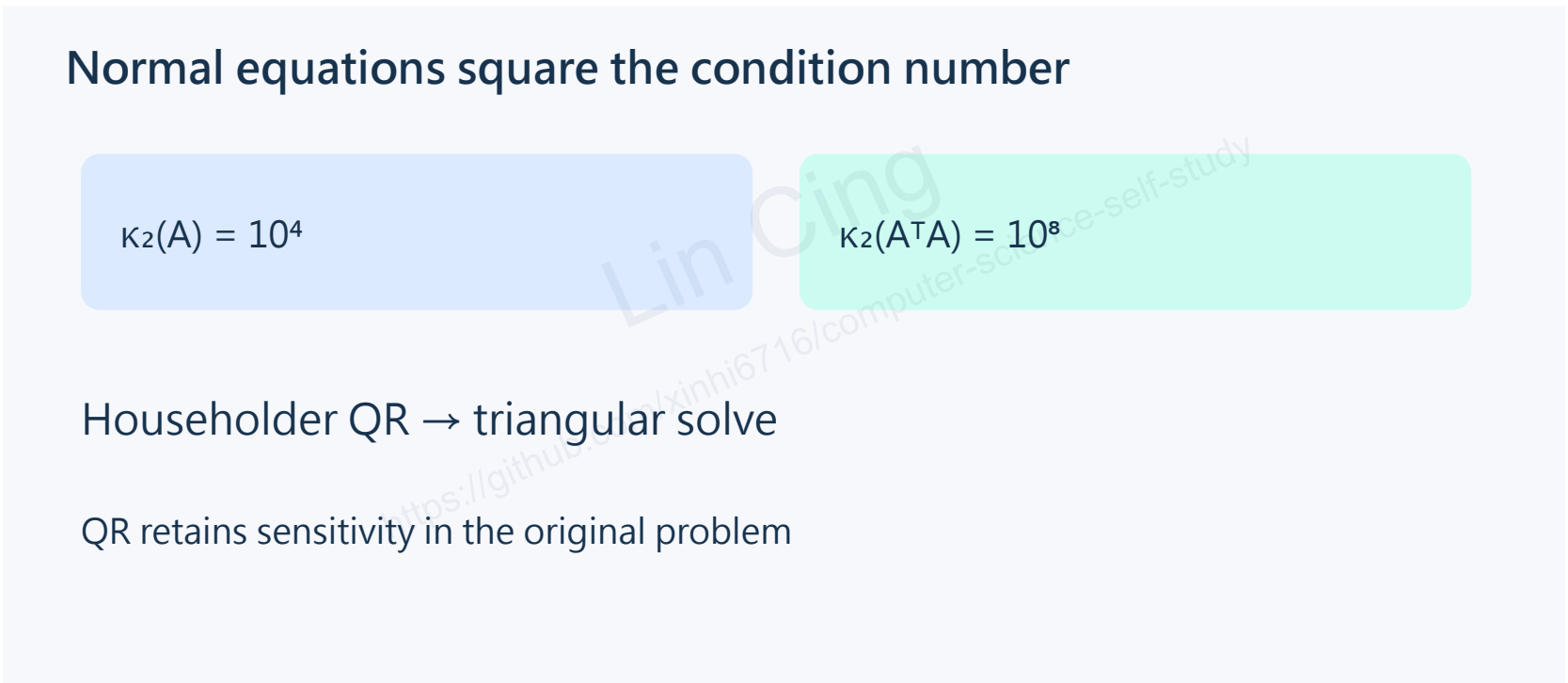
Vertical residuals and data-space orthogonality

In the scatterplot, draw vertical segments at fixed t_i with length $|b_i - \hat{b}_i|$. These are not shortest distances to the fitted line. In data space $\mathbb{R}^3$, $\hat{b} = A\hat{x} \in \text{col}(A)$ and $r \perp \text{col}(A)$. Check $A^T r = (-1/6 + 1/3 - 1/6,\ 1/3 - 2/6)^T = (0, 0)^T$.



Normal equations square the condition number

For full column rank, $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$. Since A^TA has eigenvalues σ_i^2, $\kappa_2(A^TA) = \kappa_2(A)^2$. For $A = \mathrm{diag}(1, 10^{-4})$, the numbers are 10^4 and 10^8. Forming A^TA in floating point makes weak directions harder to retain; there is no universal fixed digit loss.



QR avoids forming the squared-condition matrix

With thin QR, $A = QR$ and $Q^TQ = I$. Solve $R\hat{x} = Q^Tb$. Here $q_1 = (1,1,1)^T/\sqrt{3}$, $q_2 = (-1,0,1)^T/\sqrt{2}$, $R = [\sqrt{3}, \sqrt{3}; 0, \sqrt{2}]$, and $Q^Tb = (5/\sqrt{3}, 1/\sqrt{2})^T$, giving $(7/6, 1/2)^T$ again. Use Householder QR in practice; use pivoted QR or SVD when rank is uncertain. QR does not remove ill-conditioning in the data.

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Any rank: the normal equation is necessary

Let $f(x) = \|b - Ax\|_2^2$. If $\hat{x}$ is a global minimizer, then for every direction z, the scalar function $g(t) = f(\hat{x} + tz)$ has a minimum at $t = 0$. Thus $g'(0) = -2z^T A^T (b - A\hat{x}) = 0$.

Because this holds for every z, $A^T(b - A\hat{x}) = 0$, or $A^T A\hat{x} = A^T b$. No full-column-rank or invertibility assumption is needed.

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Any rank: the normal equation is sufficient

Conversely, suppose $A^T(b - A\hat{x}) = 0$ and write $r = b - A\hat{x}$. For any z, expand:

$$\|b - A(\hat{x} + z)\|^2 = \|r - Az\|^2 = \|r\|^2 - 2z^T A^T r + \|Az\|^2 = \|r\|^2 + \|Az\|^2 \geq \|r\|^2.$$

Therefore $\hat{x}$ is a global minimizer. Equality holds exactly when $Az = 0$, so all minimizers are $\hat{x} + \ker(A)$. Moreover $A\hat{x} \in \text{col}(A)$ and $r \perp \text{col}(A)$, making it the unique orthogonal projection of b. The projection onto a finite-dimensional column space exists, hence at least one such $\hat{x}$ exists.

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The projection matrix in the full-column-rank case

When $A \in \mathbb{R}^{m \times n}$ has full column rank, $A^T A$ is SPD and $\hat{x} = (A^T A)^{-1} A^T b$ is the unique least-squares solution. Define $P = A(A^T A)^{-1} A^T$; the fitted vector is $p = Pb$.

$P^T = P$ and $P^2 = P$, so P is the orthogonal projector onto $\mathrm{col}(A)$. Equivalently, $A^T(b - Pb) = 0$ proves the residual is orthogonal. Without full column rank $(A^T A)^{-1}$ does not exist; use the SVD/pseudoinverse formulation instead.

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Rank deficiency: the fitted vector is unique

For any real A and b, $\min_x \|Ax - b\|_2^2$ has a solution, and all solutions satisfy $A^T Ax = A^T b$. The orthogonal projection p of b onto $\mathrm{col}(A)$ exists and is unique; every x satisfying $Ax = p$ is a minimizer.

If the columns are dependent, $\ker(A)$ contains a nonzero vector. For one solution x_0, all solutions are precisely $x_0 + \ker(A)$: the fit $Ax = p$ stays unique while the coefficients need not. If $A = 0$, every x gives the same residual.

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A rank-deficient example

Take $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 0 \end{bmatrix}$, $b = (2, 3, 1)^T$, and $u = (1, 2, 0)^T$. Then $p = u(u^T b)/(u^T u) = (8/5, 16/5, 0)^T$ and $r = b - p = (2/5, -1/5, 1)^T$; check $A^T r = 0$.

$Ax = p$ means $x_1 + x_2 = 8/5$, so all solutions are $(4/5, 4/5)^T + t(1, -1)^T$. All give the same fit. The minimum-norm coefficient vector occurs at $t = 0$.

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SVD selects the unique minimum-norm solution

Let $A = U\Sigma V^T$ have nonzero singular values $\sigma_1, \ldots, \sigma_r$ and define $A^+ = \sum_{i=1}^{r} \sigma_i^{-1} v_i u_i^T$. Then $x_* = A^+ b$ is the unique minimum-norm least-squares solution; all least-squares solutions are $x_* + z$ for $z \in \ker(A)$.

Since $x_* \in \ker(A)^\perp$, $\|x_* + z\|^2 = \|x_*\|^2 + \|z\|^2$; only $z = 0$ minimizes the norm. At rank zero the empty sum gives $A^+ = 0$. Numerical implementations need a scale-aware rank cutoff.

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Why evaluate polynomials?

For coefficient vectors of lengths m and n, direct multiplication takes $O(mn)$ arithmetic operations. Evaluation at shared points turns multiplication into pointwise multiplication; interpolation returns coefficients.

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Can point values determine the coefficients?

For n points $x_0, \ldots, x_{n-1}$ in a field F, coefficient vector a and value vector y satisfy $y = Va$, where $V_{ij} = x_i^j$ for $0 \leq i, j < n$. A square matrix alone does not guarantee a unique solution.

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The Vandermonde determinant

$\det V = \prod_{0 \le i < j < n} (x_j - x_i)$. Thus V is invertible over a field exactly when the points are pairwise distinct.

Repeated points give equal rows.

For $1, 2, 4$, the determinant is $(2-1)(4-1)(4-2) = 6$. It is nonzero over the reals but zero over $\mathbb{F}_3$, where 1 and 4 coincide.

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Proof: treat the last point as a variable

Proceed by induction. For $n = 1$, the determinant and empty product both equal 1. Treat $x_0, \ldots, x_{n-2}$ as distinct indeterminates, and let $D(t)$ be the determinant whose last row is $[1, t, \ldots, t^{n-1}]$.

D has degree at most $n - 1$. Each substitution $t = x_i$ duplicates a row, so $D(x_i) = 0$. In $F[x_0, \ldots, x_{n-2}, t]$, these distinct monic linear factors are irreducible, so their product divides $D(t)$ and $D(t) = c \prod_{i=0}^{n-2} (t - x_i)$. The resulting polynomial identity can then be specialized to any field elements, including repeated points.

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Proof: identify the leading coefficient

Expand along the last row. The only term containing t^{n-1} has positive cofactor equal to the Vandermonde determinant for the first $n-1$ points.

Thus $c = \det V_{n-1} = \prod_{0 \le i < j < n-1} (x_j - x_i)$. Substitute $t = x_{n-1}$ to complete the induction. No distinctness assumption was needed in this identity.

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Lagrange interpolation constructs the solution

For pairwise distinct $x_0, \ldots, x_{n-1}$ define $L_i(z) = \prod_{j \neq i} \frac{z - x_j}{x_i - x_j}$. Each denominator is nonzero and invertible in the field. Thus $L_i(x_k) = 1$ when $i = k$ and 0 otherwise.

For arbitrary values y_i, the polynomial $P(z) = \sum_i y_i L_i(z)$ has degree less than n and $P(x_i) = y_i$. If another degree-less-than-n polynomial Q has the same values, $P - Q$ has n distinct roots and must be zero. The interpolant is unique, in agreement with invertibility of the Vandermonde matrix.

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Computing interpolating coefficients in $O(n^2)$

Construct $G(z) = \prod_{j=0}^{n-1}(z - x_j)$ by multiplying one linear factor at a time. Updating the coefficient array of length $k+1$ at step k takes $O(k)$, hence $O(n^2)$ field operations overall.

For each i, synthetic division computes $Q_i(z) = G(z)/(z - x_i)$ in $O(n)$. Horner evaluation gives $d_i = Q_i(x_i) = \prod_{j \neq i}(x_i - x_j) \neq 0$. Add the n coefficients of $y_i d_i^{-1} Q_i$ to the answer. The n iterations each cost $O(n)$, so the total is $O(n^2)$ field operations, under the unit-cost field-arithmetic model.

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Fewer than n points do not ensure uniqueness

Given $m < n$ distinct points $x_0, \ldots, x_{m-1}$, both $P(z) = 0$ and $Q(z) = \prod_{i=0}^{m-1}(z - x_i)$ vanish at all the points. They are different polynomials, both of degree less than n.

For example, to determine a polynomial of degree at most two, knowing only $P(0) = P(1) = 0$ admits both $P(z) = 0$ and $P(z) = z(z - 1)$. A value at a third distinct point permits the earlier n-point uniqueness theorem to apply.

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Horner evaluation and its loop invariant

For $A(z) = \sum_{j=0}^{n-1} a_j z^j$, fix an evaluation point x, set $b = 0$, and update $b \leftarrow a_k + xb$ for $k = n-1, n-2, \ldots, 0$. After processing k, the invariant is $b = \sum_{j=k}^{n-1} a_j x^{j-k}$. Initialization is the empty sum; each update multiplies the prior sum by x and adds a_k. At termination $k = 0$, so $b = A(x)$.

For $A(z) = 2 + 3z + 4z^2$ and $x = 2$, the successive values are 4, $3 + 2 \cdot 4 = 11$, and $2 + 2 \cdot 11 = 24$. This implementation takes $\Theta(n)$ field operations for n coefficients. Repeating it for n points costs $\Theta(n^2)$ for this method; this is not a lower bound for all multipoint evaluation methods.

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Set the transform convention

For $n = 2^s$, let $\omega_n = e^{2\pi i/n}$ and $X_k = \sum_{j=0}^{n-1} x_j \omega_n^{jk}$. This is the positive-exponent DFT; indices range from 0 to $n-1$.

$\omega_4 = i$

k: 0 1 2 3

ω_4^k: 1 i −1 −i

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Roots and periodicity

$\omega_n^n = 1$, $\omega_n^{k+n} = \omega_n^k$, and $\omega_n^{k+n/2} = -\omega_n^k$ for even n. For $n = 4$, the roots are $1, i, -1, -i$.

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Split even and odd coefficients

Write $A(z) = E(z^2) + zO(z^2)$ where $E(t) = \sum_r x_{2r} t^r$ and $O(t) = \sum_r x_{2r+1} t^r$. The two half-size DFTs reuse the same squared roots.

$$A(z) = E(z^2) + z\,O(z^2)$$

$$E: x_0, x_2, x_4, \ldots$$

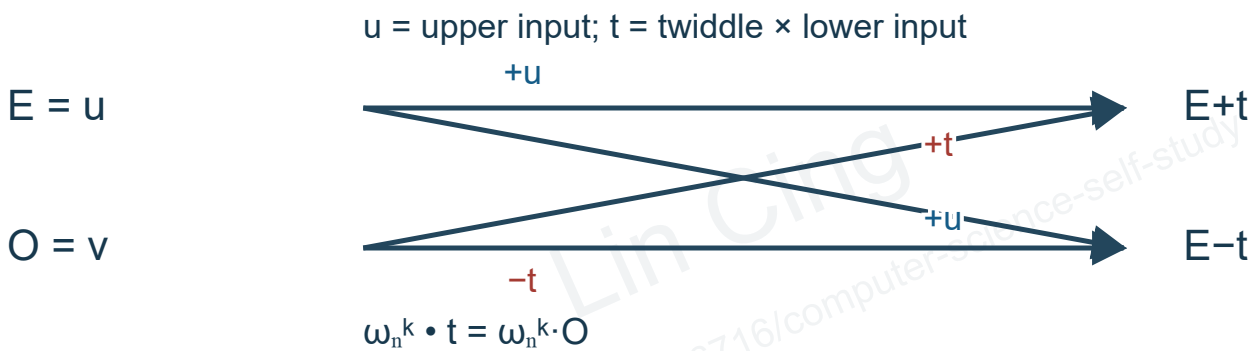
$$O: x_1, x_3, x_5, \ldots$$

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Combine a pair

For $0 \le k < n/2$, set $t = \omega_n^k O_k$. Then $X_k = E_k + t$ and $X_{k+n/2} = E_k - t$. The pair is a butterfly; the multiplier is the twiddle factor.

One butterfly



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Why the cost is $O(n \log n)$

The recurrence $T(n) = 2T(n/2) + O(n)$ has $\log_2 n$ levels and $O(n)$ work per level. The iterative in-place version has $n/2$ butterflies per stage and $\log_2 n$ stages.

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Bit-reverse the input

For $n = 4$, reverse the two-bit indices: $00, 01, 10, 11 \mapsto 00, 10, 01, 11$. Thus $[1, 2, 0, -1]$ becomes $[1, 0, 2, -1]$. The permutation prepares the iterative decimation-in-time stages.

index: 00 01 10 11

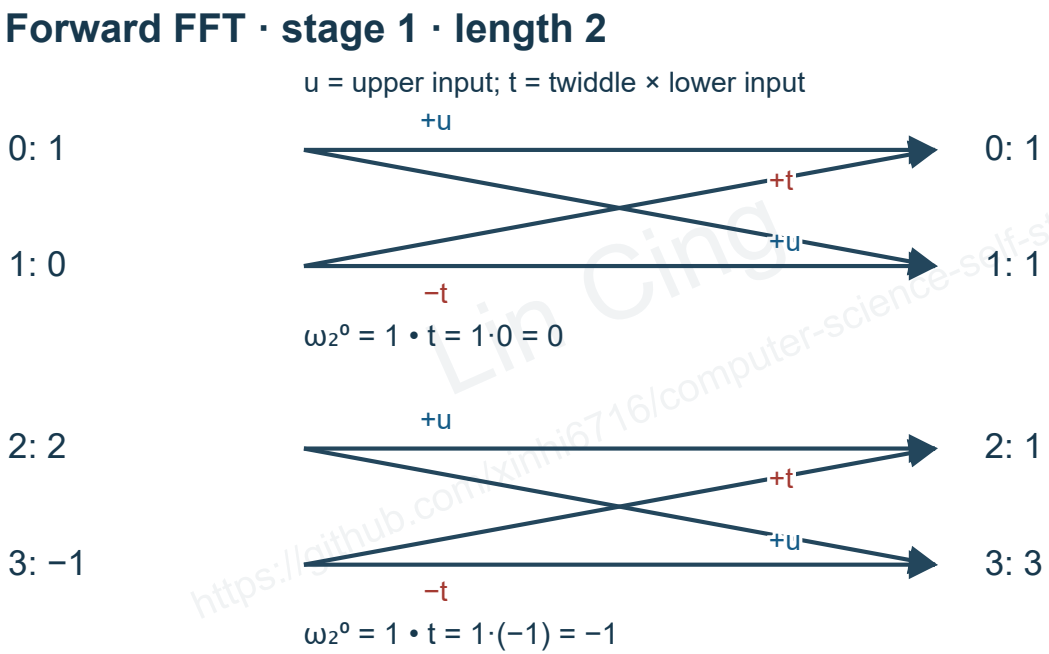
reverse: 00 10 01 11

values: 1 0 2 −1

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First stage, length 2

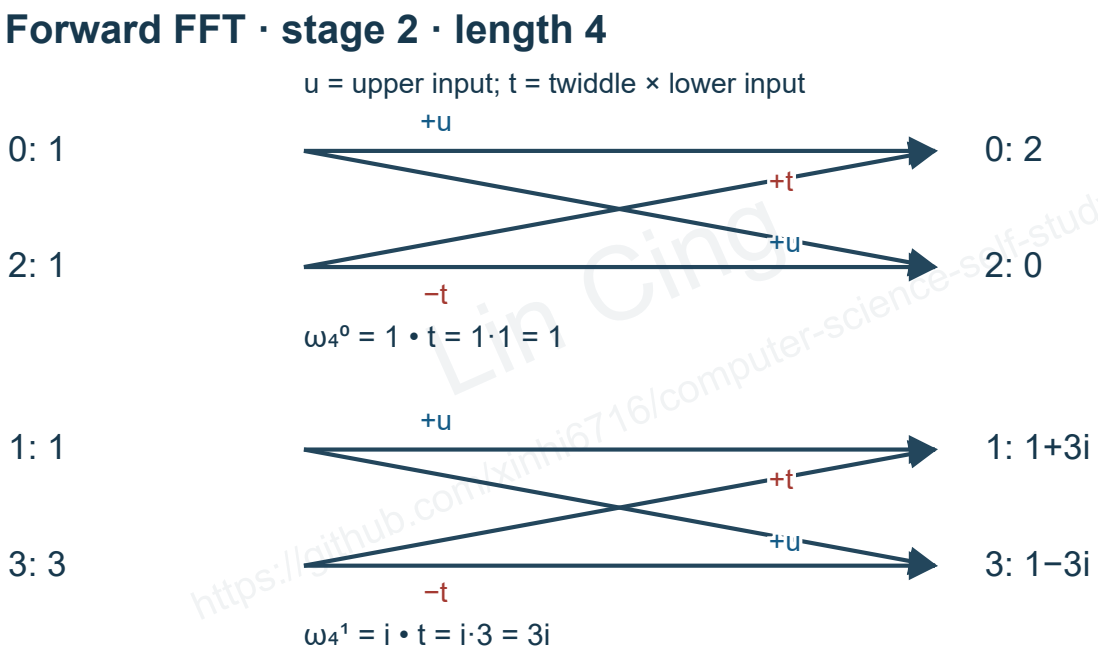
For $[1, 0, 2, -1]$, both twiddles are $\omega_2^0 = 1$. Indices $(0, 1)$ give $(1 + 0, 1 - 0) = (1, 1)$; indices $(2, 3)$ give $(2 - 1, 2 - (-1)) = (1, 3)$. State: $[1, 1, 1, 3]$.



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Second stage, length 4

Input $[1, 1, 1, 3]$. Pair $(0, 2)$ uses $\omega_4^0 = 1$: $(1 + 1, 1 - 1) = (2, 0)$. Pair $(1, 3)$ uses $\omega_4^1 = i$: $(1 + 3i, 1 - 3i)$. In index order $X = [2, 1 + 3i, 0, 1 - 3i]$.



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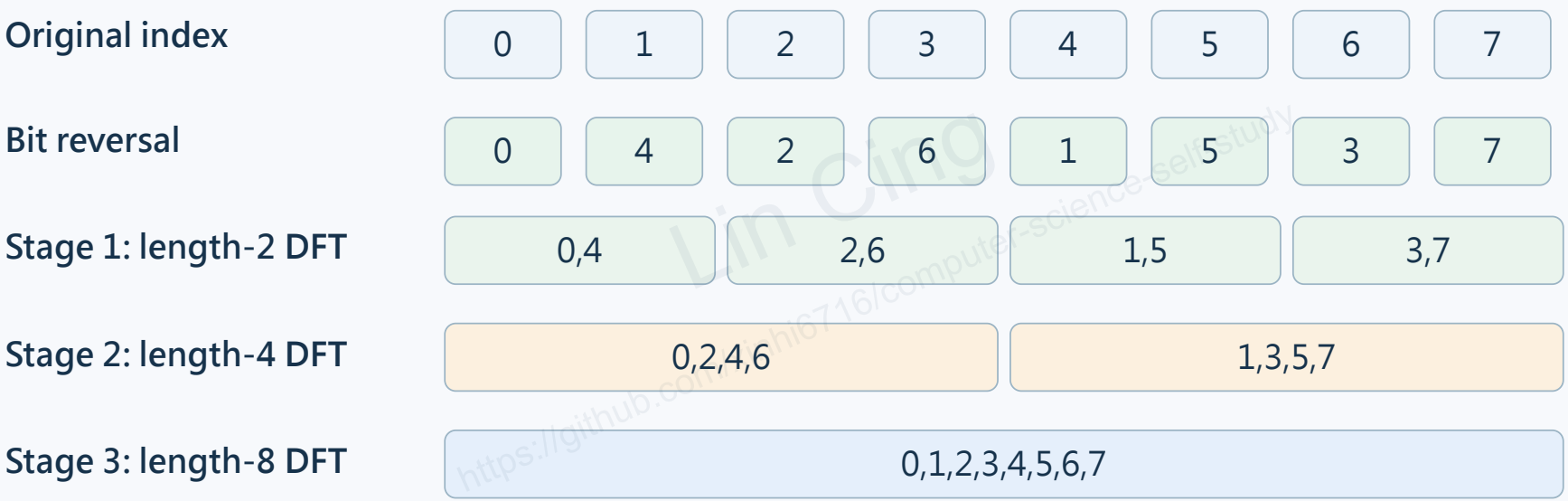
Check with the definition

Direct sums give $X_0 = 1 + 2 + 0 - 1 = 2$, $X_1 = 1 + 2i + (-1)(-i) = 1 + 3i$, $X_2 = 1 - 2 + 0 + 1 = 0$, and $X_3 = 1 - 2i + (-1)i = 1 - 3i$. This independently checks indexing and signs.

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The iterative butterfly invariant

Let $N = 2^\ell$. After stage s ($0 \le s \le \ell$), each contiguous block of length 2^s stores a length-2^s DFT in natural frequency order. Block b is the DFT of the original subsequence $x_{r+tN/2^s}$, $0 \le t < 2^s$, where r is the reversal of the $\ell - s$ bits of b.



Each block lists the original input indices; its DFT outputs use natural frequency order.

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Why each butterfly preserves the invariant

At $s = 0$, each singleton is in bit-reversed order and the claim holds. Assume it holds at stage s: the two halves of each next-stage block are DFTs of the even and odd positions of one larger subsequence. For each k, combine them as $E_k \pm \omega_{2^{s+1}}^{k} O_k$ in positions k and $k + 2^s$; the result is the DFT of that larger subsequence.

The old subsequences have stride $N/2^s$ and their starting indices differ by $N/2^{s+1}$, exactly the even/odd split of the new subsequence. Induction to $s = \ell$ gives the entire input DFT in natural frequency order.

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Bit reversal has a cost

For $N = 2^\ell$, let $\mathrm{rev}_\ell(i)$ reverse the ℓ bits of i. Scanning all ℓ bits independently for every index takes $O(N \log N)$ work. A reversed counter updates only the bits affected by a carry; over all N updates these flips total $O(N)$, so all indices can be generated in $O(N)$ work.

For in-place permutation, swap only when $i < \mathrm{rev}_\ell(i)$ so each pair is swapped once. The earlier parallel FFT assumes the map is supplied when stating $O(1)$ permutation depth; generating it must otherwise be charged separately. Word size and instruction model affect the bit-operation cost.

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The iterative algorithm

Bit-reverse x. For $L = 2, 4, \ldots, n$, for each block beginning at b and $0 \leq j < L/2$, set $u = a_{b+j}$ and $t = \omega_L^j a_{b+j+L/2}$; assign $(a_{b+j}, a_{b+j+L/2}) = (u + t, u - t)$. Save u and t before writing.

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Invert the DFT

The inverse is $x_j = \frac{1}{n} \sum_{k=0}^{n-1} X_k \omega_n^{-jk}$. Apply the same FFT with inverse roots, then divide each result by n.

bit reverse → inverse-root FFT → divide by n

[2, 1+3i, 0, 1−3i] → [1, 2, 0, −1]

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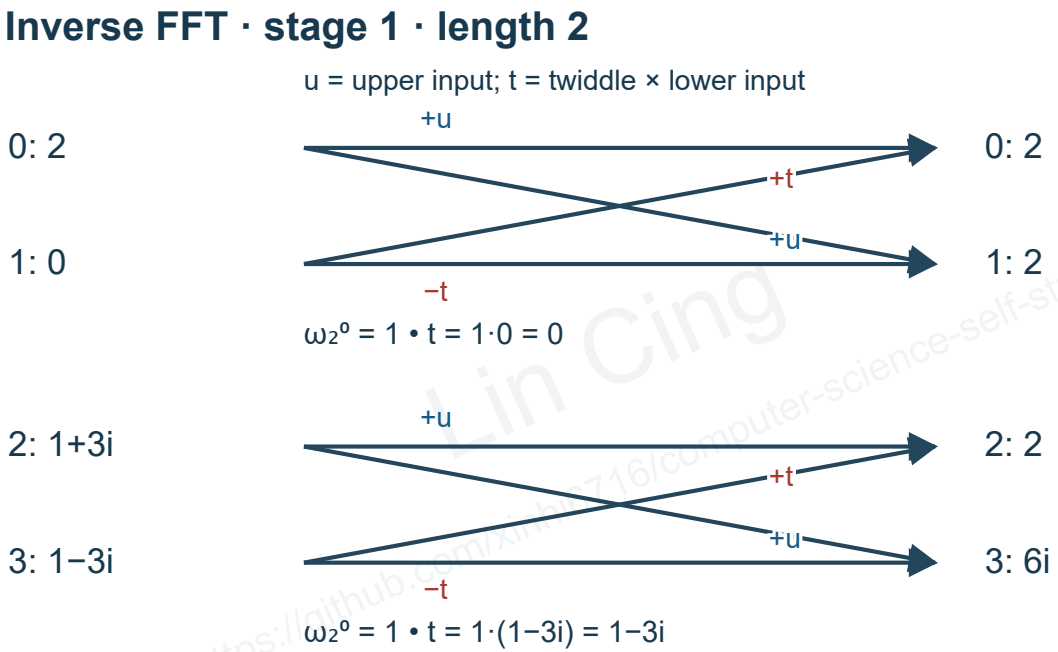
Why division by n works

Substitute X_k: $\frac{1}{n}\sum_r x_r \sum_k \omega_n^{k(r-j)}$. The inner geometric sum is n when $r = j$ and 0 otherwise because $0 \leq r, j < n$. Hence only x_j remains.

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Inverse example, first stage

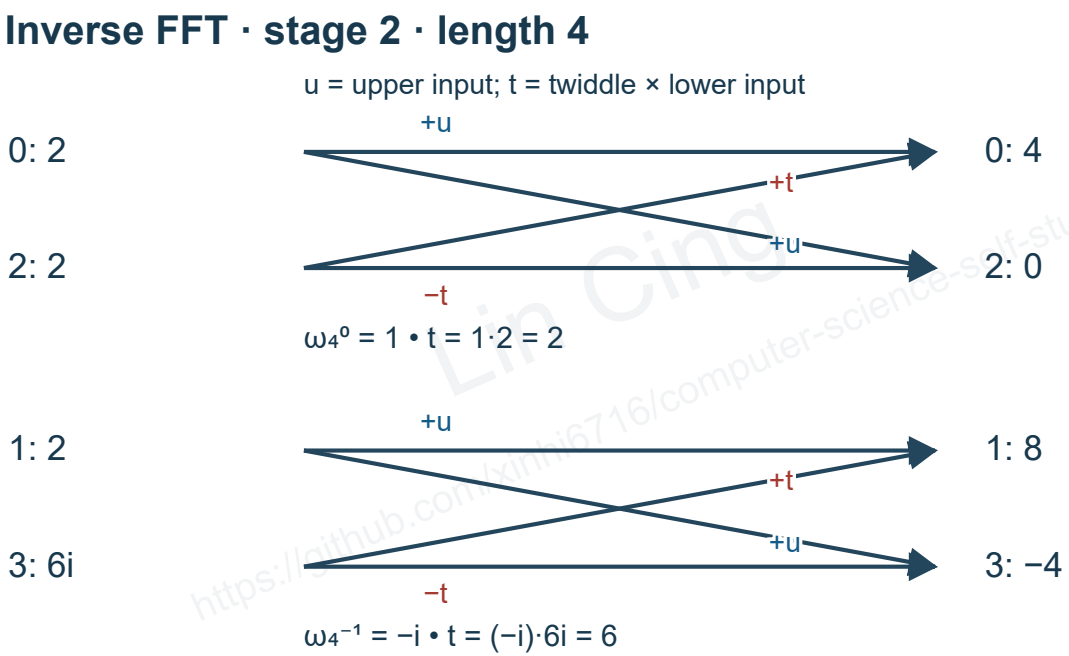
Bit-reverse $[2, 1+3i, 0, 1-3i]$ to $[2, 0, 1+3i, 1-3i]$. With length-2 twiddle 1, the pairs produce $[2, 2, 2, 6i]$.



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Inverse example, second stage

Use $\omega_4^{-1} = -i$. Pair $(0, 2)$ yields $(4, 0)$ with twiddle 1. Pair $(1, 3)$ has $(-i)(6i) = 6$ and yields $(8, -4)$. Divide $[4, 8, 0, -4]$ by 4 to recover $[1, 2, 0, -1]$.



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Recursive FFT space: count what is live at once

In a sequential depth-first implementation, suppose each level allocates temporary arrays proportional to its subproblem and releases them when finished. Along one recursion path the live capacity is at most $N + N/2 + N/4 + \cdots < 2N$; including input/output arrays and an $O(\log N)$ call stack gives $O(N)$ peak extra space.

If instead all arrays at every level are retained for tracing or reuse, each level holds N elements and there are $\log_2 N + 1$ levels, so storage is $\Theta(N \log N)$. Both approaches may perform $\Theta(N \log N)$ work; retention policy, not recursion depth alone, accounts for the space difference.

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Why applying the forward DFT twice is not inversion

Under $X_k = \sum_j x_j \omega^{jk}$, applying the same forward transform again and dividing by N gives $\frac{1}{N} \sum_k X_k \omega^{k\ell} = \frac{1}{N} \sum_j x_j \sum_k \omega^{k(j+\ell)} = x_{(-\ell) \bmod N}$.

The inner sum is N only when $j + \ell \equiv 0 \pmod N$ and is zero otherwise. Thus the result reverses cyclic indices. The inverse instead uses ω^{-1} and divides by N, or follows the second forward transform with index reversal.

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A concrete counterexample

Let $N = 4$, $\omega = i$ and $x = [1, 2, 0, 0]$. The first forward DFT is $[3, 1 + 2i, -1, 1 - 2i]$. Applying the same forward root a second time and dividing by 4 produces $[1, 0, 0, 2]$, not the input $[1, 2, 0, 0]$. The issue is the direction of the exponent, not a missing factor of N; using inverse root $-i$ restores the input.

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Work and depth of parallel FFT

Use a synchronous random-access EREW PRAM, with $N = 2^s \geq 2$ and a given bit-reversal address map. Assign $N/2$ processors per stage. Each processor reads one disjoint input pair, computes one butterfly and writes the pair back; distinct butterflies access disjoint locations. A barrier separates stages. Assume twiddle factors are precomputed.

Bit-reversal copying takes $O(N)$ work and $O(1)$ time; each of $\log_2 N$ stages takes $\Theta(N)$ work and $\Theta(1)$ time. Therefore $W = \Theta(N \log N)$, depth $D = \Theta(\log N)$ and extra storage $O(N)$. This is a model result, not a promise for every machine. Charge index or twiddle generation separately if they are not supplied.

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Linear versus circular convolution

Pointwise multiplying length-N transforms gives circular convolution $c_k = \sum_{j=0}^{N-1} a_j b_{(k-j) \bmod N}$. Without enough padding, coefficients wrap and add to earlier indices.

$$m + n - 1 \le N$$

$$[1, 2, 0, 0] \times [3, 4, 0, 0]$$

linear output: [3, 10, 8, 0]

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Pad before multiplying

For lengths $m, n \geq 1$, choose a power of two $N \geq m + n - 1$, append zeros to both inputs, FFT, multiply pointwise, and inverse FFT. The first $m + n - 1$ outputs are the linear convolution.

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Convolution example

$[1, 2] * [3, 4] = [3, 10, 8]$. With $N = 4$, pad to $[1, 2, 0, 0]$ and $[3, 4, 0, 0]$. The inverse pointwise product gives $[3, 10, 8, 0]$; $N = 2$ would instead wrap 8 into the constant term.

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Complex FFT convolution: two forward transforms

For $A(z) = 1 + 2z$, $B(z) = 3 + 4z$, the linear convolution has length $2 + 2 - 1 = 3$, so pad to $N = 4$. Use the positive-exponent convention $\omega = i$: $A = [1, 2, 0, 0]$, $B = [3, 4, 0, 0]$.

Evaluate $X_k = \sum_j x_j i^{jk}$ at $1, i, -1, -i$: $\widehat{A} = [3, 1 + 2i, -1, 1 - 2i]$ and $\widehat{B} = [7, 3 + 4i, -1, 3 - 4i]$. For example, $A(i) = 1 + 2i$ and $B(-1) = 3 - 4 = -1$.

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Multiply matching frequencies and invert

The pointwise products are $C = [21, -5 + 10i, 1, -5 - 10i]$; for example $(1 + 2i)(3 + 4i) = -5 + 10i$. The inverse uses $i^{-1} = -i$: $c_j = \frac{1}{4} \sum_{k=0}^{3} C_k (-i)^{jk}$.

The four unscaled sums are $[12, 40, 32, 0]$, so $c = [3, 10, 8, 0]$. Direct multiplication confirms $(1 + 2z)(3 + 4z) = 3 + 10z + 8z^2$. Pair values at the same frequency, not across frequencies.

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Multiply matching frequencies and invert (continued)

Frequency	Value of A	Value of B		Pointwise product
1	3	7	×	21
i	1+2i	3+4i	×	−5+10i
−1	−1	−1	×	1
−i	1−2i	3−4i	×	−5−10i

Inverse: [21, −5+10i, 1, −5−10i] → [3, 10, 8, 0]

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Fourier transforms over a finite field

Choose a prime p, let $N = 2^s$ and $N \mid (p-1)$. Choose $\omega \in \mathbb{F}_p$ of multiplicative order exactly N: $\omega^N = 1$ and $\omega^k \neq 1$ for $1 \leq k < N$. Such a root exists because the nonzero elements form a cyclic group of order $p-1$.

Define $X_k = \sum_{j=0}^{N-1} x_j \omega^{jk} \pmod{p}$; all arithmetic takes place in $\mathbb{F}_p$.

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Why the inverse works

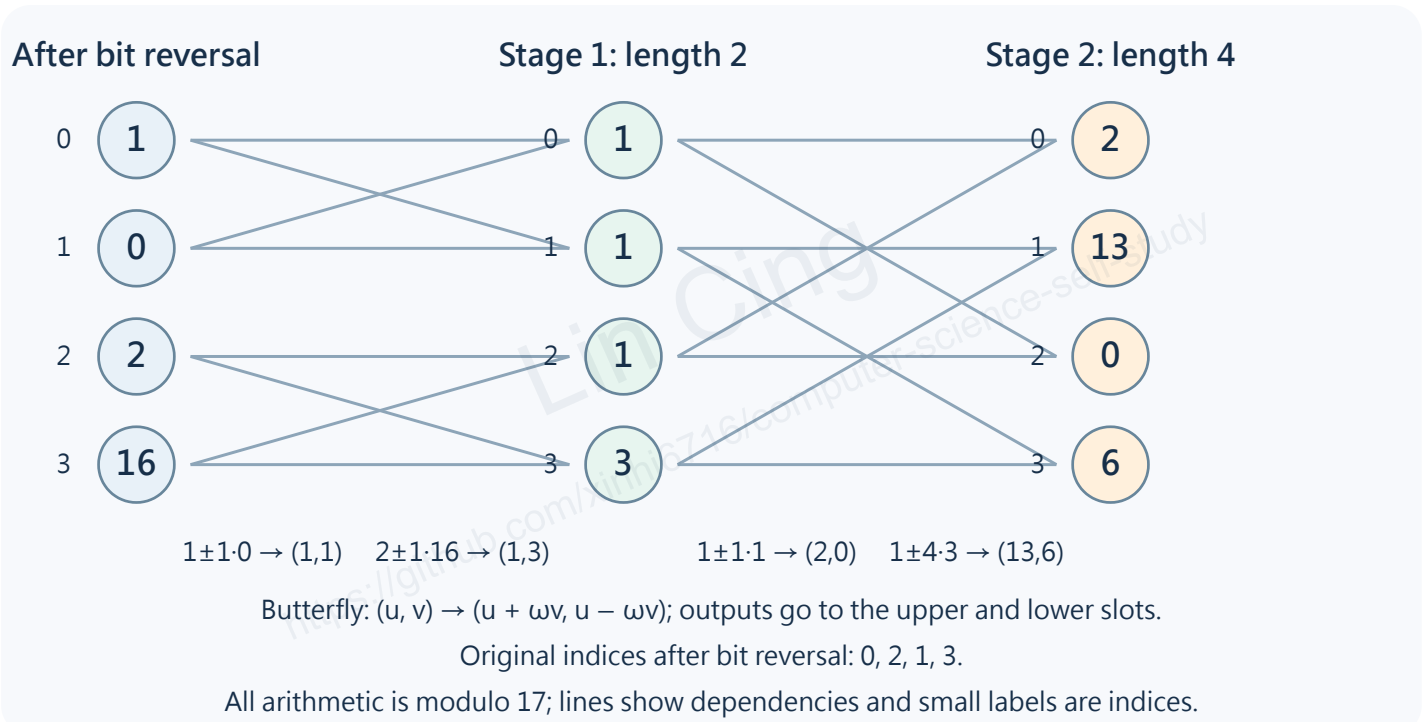
Since $N \mid p-1$, $p \nmid N$, so N^{-1} exists. If $0 < r < N$, then $\omega^r \neq 1$ and $1-\omega^r$ is invertible; consequently $\sum_{k=0}^{N-1} \omega^{kr} = 0$. For $r = 0$, the sum is N.

Substitution and exchange of sums give $x_j = N^{-1} \sum_{k=0}^{N-1} X_k \omega^{-jk} \pmod{p}$. The proof uses invertibility of nonzero field elements; it does not automatically extend to composite moduli.

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NTT example: two butterfly stages modulo 17

Take $N = 4, p = 17, \omega = 4$. Since $4^2 = 16 = -1$ and $4^4 = 1$, its order is exactly 4; $4^{-1} = 13$. Input $[1, 2, 0, -1]$ is $[1, 2, 0, 16]$ modulo 17.



Direct evaluation also gives $[2, 13, 0, 6]$. Interpret every negative value modulo 17.

Pointwise product and integer recovery

Pad $A = [1, 2]$ and $B = [3, 4]$ to $N = 4$. Their NTTs are $[3, 9, 16, 10]$ and $[7, 2, 16, 4]$; their pointwise product is $[4, 1, 1, 6]$. The inverse is $[3, 10, 8, 0]$, giving $3 + 10z + 8z^2$.

Linear rather than cyclic convolution requires $N \geq r + s - 1$. A modular result identifies only a residue. If every integer coefficient has known absolute bound M and $p > 2M$, its symmetric representative is unique. Here $r = s = 2$, $|a_i|, |b_i| \leq 4$ gives the coarse bound $M = 2 \cdot 4^2 = 32$, but $17 \not> 64$; the residue 10 also represents -7, so without a sufficiently tight range bound the integer coefficient cannot be identified uniquely.

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Signed integer convolution with a sufficient bound

Take $A = [1, 2]$ and $B = [1, -1]$, again with $p = 17$, $N = 4$, $\omega = 4$. Their padded transforms are $[3, 9, 16, 10]$ and $[0, 14, 2, 5]$. The pointwise product $[0, 7, 15, 16]$ inverses to $[1, 1, 15, 0]$.

Both inputs have coefficient magnitude at most $B = 2$; each convolution coefficient contains at most $\min(2, 2) = 2$ products, so $M = 2B^2 = 8$. Since $p = 17 > 2M = 16$, the symmetric representatives in $[-8, 8]$ are unique: residue 15 decodes to -2, giving $[1, 1, -2, 0]$.

A length-N FFT uses $\Theta(N \log N)$ arithmetic operations modulo p. If p grows with the input, the bit cost of modular additions and multiplications depends on the arithmetic method for $\log p$-bit integers; it is not automatically constant.

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Precision and cost

Three transforms and N pointwise products take $O(N \log N)$ arithmetic and $O(N)$ storage. This is an arithmetic-operation bound, not a bit-complexity guarantee. Floating-point roundoff accumulates; rounding to integer coefficients is valid only when every absolute error is proved below $1/2$. Otherwise use higher precision or an exact method.

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Choose the right method

For small inputs direct multiplication may be simpler. FFT requires suitable padding and a numerical error budget; for exact integer results, consider number-theoretic transforms with suitable moduli and reconstruction bounds. Check convention, indices, and normalization at interfaces.

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References and attribution

CLRS, Introduction to Algorithms, third edition, chapters 28 and 30. Matrix: PDF pages 834–861. FFT: PDF pages 919, 921, 923, 928–936, 939–941, 944–945 (PDF page numbering, third edition). MIT 6.046J Fall 2004 lecture 3 slides, PDF pages 27 and 35–43, and Fall 2005 lecture 3 notes support matrix multiplication and Strassen, not FFT.

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